

Chapter Summary

Key Terms

2.1 Statements and Quantifiers

- logic
- logical statement
- truth values
- symbolic form
- negation of a logical statement
- quantifier
- premises
- conclusion
- inductive logical arguments

2.2 Compound Statements

- compound statement
- connective
- conjunction
- disjunction
- conditional
- hypothesis
- conclusion
- biconditional
- dominance of connectives

2.3 Constructing Truth Tables

- Truth table
- Multiplication principle
- Valid

2.5 Equivalent Statements

- logically equivalent
- tautology
- inverse
- converse
- contrapositive

2.6 De Morgan's Laws

- Boolean logic
- negation of a conditional

2.7 Logical Arguments

- sound
- fallacy
- deductive arguments
- law of detachment
- law of denying the consequent
- chain rule for conditional arguments

Key Concepts

2.1 Statements and Quantifiers

- Logical statements have the form of a complete sentence and make claims that can be identified as true or false.
- Logical statements are represented symbolically using a lowercase letter.
- The negation of a logical statement has the opposite truth value of the original statement.
- Be able to
 - Determine whether a sentence represents a logical statement.

- Write and translate logical statements between words and symbols.
- Negate logical statements, including logical statements containing quantifiers of *all*, *some*, and *none*.

2.2 Compound Statements

- Logical connectives are used to form compound logical statements by using words such as *and*, *or*, and *if ..., then*.
- A conjunction is a compound logical statement formed by combining two statements with the words “and” or “but.” If the two independent clauses are represented by p and q , respectively, then the conjunction is written symbolically as $p \wedge q$. For the conjunction to be true, both p and q must be true.
- A disjunction joins two logical statements with the *or* connective. In logic *or* is inclusive. For an *or* statement to be true at least one statement must be true, but both may also be true.
- A conditional statement has the form if p , then q , where p and q are logical statements. The only time the conditional statement is false is when p is true, and q is false.
- The biconditional statement is formed using the connective if and only if for the biconditional statement to be true, the true values of p and q , must match. If p is true then q must be true, if p is false, then q must be false.
- Translate compound statements between words and symbolic form.

Connective	Symbol	Name
and but	$\wedge$	conjunction
or	$\vee$	disjunction, inclusive or
not	$\sim$	negation
if ..., then implies	$\rightarrow$	conditional, implication
if and only if	$\leftrightarrow$	biconditional

- The dominance of connectives explains the order in which compound logical statements containing multiple connectives should be interpreted.
- The dominance of connectives should be applied in the following order
 - Parentheses
 - Negations
 - Disjunctions/Conjunctions, left to right
 - Conditionals
 - Biconditionals

Dominance	Connective	Symbol	Evaluate
Least Dominant	Parentheses	()	First ↓ Left to right or add parentheses to specify order because or/and have equal dominance. ↓ Last
	Negation	$\sim$	
	Disjunction/Conjunction	$\vee, \wedge$	
	Conditional	$\rightarrow$	
Most Dominant	Biconditional	$\leftrightarrow$	

Figure 2.18

2.3 Constructing Truth Tables

- Determine the true values of logical statements involving negations, conjunctions, and disjunctions.
 - The negation of a logical statement has the opposite true value of the original statement.
 - A conjunction is true when both p and q are true, otherwise it is false.

- A disjunction is false when both p and q are false, otherwise it is true.
- Know how to construct a truth table involving negations, conjunctions, and disjunctions and apply the dominance of connectives to determine the truth value of a compound logical statement containing, negations, conjunctions, and disjunctions.

Negation		Conjunction (AND)			Disjunction (OR)		
p	$\sim p$	p	q	$p \wedge q$	p	q	$p \vee q$
T	F	T	T	T	T	T	T
F	T	T	F	F	T	F	T
		F	T	F	F	T	T
		F	F	F	F	F	F

- A logical statement is valid if it is always true. Know how to construct a truth table for a compound statement and use it to determine the validity of compound statements involving negations, conjunctions, and disjunctions.

2.4 Truth Tables for the Conditional and Biconditional

- The conditional statement, if p then q , is like a contract. The only time it is false is when the contract has been broken. That is, when p is true, and q is false.

Conditional		
p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- The biconditional statement, p if and only if q , is true whenever p and q have matching true values, otherwise it is false.

Biconditional		
p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- Know how to construct truth tables involving conditional and biconditional statements.
- Use truth tables to analyze conditional and biconditional statements and determine their validity.

2.5 Equivalent Statements

- Two statements p and q are logically equivalent if the biconditional statement, $p \leftrightarrow q$ is a valid argument. That is, the last column of the truth table consists of only true values. In other words, $p \leftrightarrow q$ is a tautology. Symbolically, p is logically equivalent to q is written as: $p \equiv q$.
- A logical statement is a tautology if it is always true.
- To be valid a local argument must be a tautology. It must always be true.
- Know the variations of the conditional statement, be able to determine their truth values and compose statements with them.
- The converse of a conditional statement, if p then q , is the statement formed by interchanging the hypothesis and conclusion. It is the statement if q then p .
- The inverse of a conditional statement if formed by negating the hypothesis and the conclusion of the conditional statement.
- The contrapositive negates and interchanges the hypothesis and the conclusion.

				Conditional	Contrapositive	Converse	Inverse
p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim q \rightarrow \sim p$	$q \rightarrow p$	$\sim p \rightarrow \sim q$
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	T
F	T	T	F	T	T	F	F
F	F	T	T	T	T	T	T

- The conditional statement is logically equivalent to the contrapositive.
- The converse is logically equivalent to the inverse.
- Know how to construct and use truth tables to determine whether statements are logically equivalent.

2.6 De Morgan's Laws

- De Morgan's Law for the negation of a disjunction states that, $\sim(p \vee q)$ is logically equivalent to $\sim p \wedge \sim q$.
- De Morgan's Law The negation of a conjunction states that, $\sim(p \wedge q) \equiv \sim p \vee \sim q$.
- Use De Morgan's Laws to negate conjunctions and disjunctions.
- The negation of a conditional statement, if p then q is logically equivalent to the statement p and not q . Use this property to write the negation of conditional statements.
- Use truth tables to evaluate De Morgan's Laws.

2.7 Logical Arguments

- A logical argument uses a series of facts or premises to justify a conclusion or claim. It is valid if its conclusion follows from the premises, and it is sound if it is valid, and all of its premises are true.
- The law of detachment is a valid form of a conditional argument that asserts that if both the conditional, $p \rightarrow q$ is true and the hypothesis, p is true, then the conclusion q must also be true.

Law of Detachment	
Premise:	$p \rightarrow q$
Premise:	p
Conclusion:	$\therefore q$

- Know how to apply the law of detachment to determine the conclusion of a pair of statements.
- The law of denying the consequent is a valid form of a conditional argument that asserts that if both the conditional, $p \rightarrow q$ is true and the negation of the conclusion, $\sim q$ is true, then the negation of the hypothesis $\sim p$

must also be true.

Law of Denying the Consequent	
Premise:	$p \rightarrow q$
Premise:	$\sim q$
Conclusion:	$\therefore \sim p$

- Know how to apply the law of denying the consequent to determine the conclusion for pairs of statements.
- The chain rule for conditional arguments is a valid form of a conditional argument that asserts that if the premises of the argument have the form, $p \rightarrow q$ and $q \rightarrow r$, then it follows that $p \rightarrow r$.

Chain Rule for Conditional Arguments	
Premise:	$p \rightarrow q$
Premise:	$q \rightarrow r$
Conclusion:	$\therefore p \rightarrow r$

- Know how to apply the chain rule to determine valid conclusions for pairs of true statements.

Videos

2.1 Statements and Quantifiers

- [Logic Part 1A: Logic Statements and Quantifiers \(https://openstax.org/r/Logic_Statements_and_Quantifiers\)](https://openstax.org/r/Logic_Statements_and_Quantifiers)

2.2 Compound Statements

- [Logic Part 1B: Compound Statements, Connectives and Symbols \(https://openstax.org/r/Compound_Statements\)](https://openstax.org/r/Compound_Statements)

2.3 Constructing Truth Tables

- [Logic Part 2: Truth Values of Conjunctions: Is an "AND" statement true or false? \(https://openstax.org/r/Truth_Values_of_Conjunctions\)](https://openstax.org/r/Truth_Values_of_Conjunctions)
- [Logic Part 3: Truth Values of Disjunctions: Is an "OR" statement true or false? \(https://openstax.org/r/Truth_Values_of_Disjunctions\)](https://openstax.org/r/Truth_Values_of_Disjunctions)
- [Logic Part 4: Truth Values of Compound Statements with "and", "or", and "not" \(https://openstax.org/r/opL9I4tZCC0\)](https://openstax.org/r/opL9I4tZCC0)
- [Logic Part 5: What are truth tables? How do you set them up? \(https://openstax.org/r/-tdSRqLGhaw\)](https://openstax.org/r/-tdSRqLGhaw)
- [Logic Part 6: More on Truth Tables and Setting Up Rows and Column Headings \(https://openstax.org/r/j3kKnUNIT6c\)](https://openstax.org/r/j3kKnUNIT6c)

2.4 Truth Tables for the Conditional and Biconditional

- [Logic Part 8: The Conditional and Tautologies \(https://openstax.org/r/Conditional_and_Tautologies\)](https://openstax.org/r/Conditional_and_Tautologies)
- [Logic Part 11B Biconditional and Summary of Truth Value Rules in Logic \(https://openstax.org/r/omKzui0Fytk\)](https://openstax.org/r/omKzui0Fytk)
- [Logic Part 13: Truth Tables to Determine if Argument is Valid or Invalid \(https://openstax.org/r/AQB3svnxxiw\)](https://openstax.org/r/AQB3svnxxiw)

2.7 Logical Arguments

- [Logic Part 14: Common Argument Forms like Modus Ponens and Tollens \(https://openstax.org/r/Modus_Ponens_and_Tollens\)](https://openstax.org/r/Modus_Ponens_and_Tollens)

Projects

Logic Gates

Logic gates are the basis for all digital circuits.

1. Research and document the following terms: logic gate, OR gate, AND gate, and NOT gate.
2. Construct a diagram of a NAND gate, NOR gate, and a XOR gate by using at least two of the following gates: AND, OR, and NOT.

3. Digital electronics use a 1 for true or on, and a 0 for false or off. Create a truth table documenting all possible cases using 0s and 1s for the NAND gate, NOR gate and XOR gate.
4. Use a truth table to explain how XOR is related to the biconditional statement.

Logical Fallacies

Fallacies are false or deceptive logical arguments.

1. Research and document the structure of five of the following named fallacies: hasty generalization, limited choice, false cause, appeal to popularity, appeal to emotion, appeal to authority, personal attack, gamblers' ruin, slippery slope, and circular reasoning.
2. Create a presentation highlighting one of the five fallacies researched in the previous question. The presentation must include an introductory slide with the title of the fallacy and the form or structure of the argument. The second slide must include an example of this fallacy as used in a commercial, a political cartoon or a current event or new article. The third slide must include an explanation of why the example on slide to is a representative example of the fallacy. The last slide must include citations for any materials used. No textbooks should be used as reference.

Careers in Logic

Lawyers, mathematicians, and computer programmers are a few of the careers that require knowledge of logic.

1. What career are you interested in? Research how knowledge of logic applies to your chosen field of study. Then, write a cover letter for a position in your field you'd like to apply to. In the cover letter, include how your knowledge of logic qualifies you for the position you are applying for. If you do not think logic is important for your given career choice, find a position where logic is an essential element of the position and complete the project by pretending you are writing a cover letter for that job.

Chapter Review

Statements and Quantifiers

Fill in the blanks to complete the following sentences.

1. The _____ of a logical statement has the opposite truth value of the original statement.
2. _____ are logical statements presented as the facts used to support the conclusion of a logical argument.

Determine whether each of the following sentences represents a logical statement, also called a proposition. If it is a logical statement, determine whether it is true or false.

3. Where is the restroom?
4. No even numbers are odd numbers.
5. $4 + 3 = 8$.

Write the negation of each following statement symbolically and in words.

6. $\sim p$: Pink Floyd's album *The Wall* is not a rock opera.
7. q : Some dogs are Labrador retrievers.
8. $\sim r$: Some universities are not expensive.

Draw a logical conclusion to the following arguments, and include in both one of the following quantifiers: all, some, or none.

9. Spaghetti noodles are made with wheat, ramen noodles are made with wheat, and lo mein noodles are made with wheat.
10. A Porsche Boxster does not have four doors, a Volkswagen Beetle does not have four doors, and a Mazda Miata does not have four doors.

Compound Statements

Fill in the blanks to complete the following sentences.

11. _____ are words or symbols used to join two or more logical statement together to form a compound statement.
12. _____ and _____ have equal dominance and are evaluated from left to right when no parentheses are present in a compound logical statement.

Translate each compound statement below into symbolic form.

Given: p : "Tweety Bird is a bird," q : "Bugs is a bunny," r : "Bugs says, 'What's up, Doc?'," s : "Sylvester is a cat," and t : "Sylvester chases Tweety Bird."

13. If Tweety Bird is a bird, then Sylvester will not chase him.
14. Tweety Bird is a bird and Sylvester chases him if and only if Bugs says, "What's up Doc?"

Translate the symbolic form of each compound logical statement below into words.

Given: p : "Tweety Bird is a bird," q : "Bugs is a bunny," r : "Bugs says, 'What's up, Doc?'," s : "Sylvester is a cat," and t : "Sylvester chases Tweety Bird."

15. $\sim q \vee p \rightarrow \sim s$
16. $\sim(p \wedge \sim s) \leftrightarrow q \rightarrow r$

For each of the following compound logical statements, apply the proper dominance of connectives by adding parentheses to indicate the order to evaluate the statement.

17. $p \vee q \wedge r \rightarrow \sim s \wedge t$
18. $\sim p \rightarrow q \vee r \leftrightarrow p \wedge s \rightarrow \sim t$

Constructing Truth Tables

Fill in the blanks to complete the sentences.

19. A _____ is true if at least one of its component statements is true.
20. For a _____ to be true, all of its component statements must be true.

Given the statements, p : "No fish are mammals," q : "All lions are cats," and $\sim r$: "Some birds do not lay eggs," construct a truth table to determine the truth value of each compound statement below.

21. $p \wedge \sim r$
22. $\sim(p \vee \sim r)$
23. $\sim p \vee q \wedge \sim(\sim r)$

Construct a truth table to analyze all the possible outcomes of the following statements, and determine whether the statements are valid.

24. $\sim p \vee q \wedge p$

25. $\sim p \vee q \vee \sim q$

Truth Tables for the Conditional and Biconditional

Fill in the blanks to complete the following sentences.

26. If the _____, p , of a conditional statement is true, then the conclusion, q , must also be true for the conditional statement $p \rightarrow q$ to be true.
27. The biconditional statement $p \leftrightarrow q$ is _____ whenever the truth value of p matches the truth value of q , otherwise it is _____.

Complete the truth tables below to determine the truth value of the proposition in the last column.

28.

p	q	r	$p \vee q$	$\sim(p \vee q)$	$\sim r$	$\sim(p \vee q) \rightarrow \sim r$
F	F	T				

29.

p	q	$\sim q$	$p \rightarrow q$	$\sim(p \rightarrow q)$	$p \wedge \sim q$	$\sim(p \rightarrow q) \leftrightarrow (p \wedge \sim q)$
T	F					

Assume the following statements are true. p : "Poof is a baby fairy," q : "Timmy Turner has fairly odd parents," r : "Cosmo and Wanda will grant Timmy's wishes," and t : "Timmy Turner is 10 years old." Translate each of the following statements into symbolic form, then determine its truth value.

30. If Timmy Turner is 10 years old and Poof is not a baby fairy, then Timmy Turner has fairly odd parents.
31. Cosmo and Wanda will not grant Timmy's wishes if and only if Timmy Turner is 10 years old or he does not have fairly odd parents.
32. Construct a truth table to analyze all the possible outcomes and determine the validity of the following argument.
- $$\sim p \vee q \leftrightarrow \sim q \rightarrow \sim p$$

Equivalent Statements

Fill in the blanks to complete the sentences below.

33. The _____ is logically equivalent to the inverse $\sim p \rightarrow \sim q$.
34. The _____ is logically equivalent to the conditional $p \rightarrow q$.

Use the conditional statement, $p \rightarrow q$: "If Novak makes the basket, then Novak's team will win the game," to answer the following questions.

35. Write the conclusion of the conditional statement in words and label it appropriately.
36. Write the hypothesis of the conditional statement in words and label it appropriately.
37. Identify the following statement as the converse, inverse, or contrapositive: "If Novak does not make the basket, then his team will not win the game."
38. Identify the following statement as the converse, inverse, or contrapositive: "If Novak's team wins the game, then he made the basket."

De Morgan's Laws

Fill in the blanks to complete the sentences.

39. De Morgan's Law for the negation of a disjunction states that $\sim(p \vee q) \equiv$ _____.
40. De Morgan's Law for the negation of a conjunction states that _____ $\equiv \sim p \vee \sim q$.
41. Apply De Morgan's Law to write the statement without parentheses: $\sim(\sim p \wedge q)$.
42. Apply the property for the negation of a conditional to write the statement as a conjunction or disjunction: $\sim(\sim p \wedge q \rightarrow \sim r)$.
43. Write the negation of the conditional statement in words: If Thomas Edison invented the phonograph, then albums are made of vinyl, or the transistor radio was the first portable music device.
44. Construct a truth table to verify that the logical property is valid: $\sim(\sim p \rightarrow \sim q) \equiv \sim p \wedge q$.

Logical Arguments

Fill in the blanks to complete the sentences below.

45. The _____ is a valid logical argument with premises, $p \rightarrow q$ and p , used to support the conclusion, q .
46. The chain rule for conditional arguments states that the _____ property applies to conditional arguments, so that: $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$.

Assume each pair of statements represents true premises in a logical argument. Based on these premises, state a valid conclusion that is consistent with the form of the argument.

47. If the Tampa Bay Buccaneers did not win Super Bowl LV, then Tom Brady was not their quarterback. Tom Brady was the Tampa Bay Buccaneers quarterback.
48. If $\sim q$, then p and if r , then $\sim q$.
49. If Kamala Harris is the vice president of the United States, then Kamala Harris is the president of the U.S. Senate. Kamala Harris is the vice president of the United States.
50. Construct a truth table or Venn diagram to prove whether the following argument is valid. If the argument is valid, determine whether it is sound.
If all frogs are brown, then Kermit is not a frog. Kermit is a frog. Therefore, some frogs are not brown.

Chapter Test

Determine whether each of the following sentences represent a logical statement. If it is a logical statement, determine whether it is true or false.

1. $1 + 2 - 3 = 0$
2. Please, sit down over there.
3. All mammals lay eggs.

Write the negation of each statement below.

4. Some monkeys do not have tails.
5. $p \wedge \sim q$
6. If the plumber does not remove the clog, then the homeowner will not pay the plumber.

Given: p : Frodo is a hobbit, q : Gandalf is a wizard, r : Frodo and Samwise will take the ring to Mordor, and s : Gollum will help Frodo get into Mordor.

Translate the symbolic form of each compound logical statement into words.

7. $p \vee q \leftrightarrow r$
8. $\sim(\sim r \vee \sim s)$

Translate the written form of each compound logical statement into symbolic form.

9. Frodo and Samwise will take the ring to Mordor or Gandalf is not a wizard and Frodo is a hobbit.
10. If Gollum will not help Frodo get into Mordor, then Gandalf is not a wizard and Frodo is not a hobbit.

For each of the following compound logical statements, apply the proper dominance of connectives by adding parentheses to indicate the order in which the statement must be evaluated.

11. $\sim p \rightarrow q \leftrightarrow r$
12. $\sim(p \wedge q) \leftrightarrow \sim p \vee \sim q$

13. Complete the truth table to determine the truth value of the proposition in the last column.

p	q	$p \wedge q$	$\sim q$	$p \vee \sim q$	$(p \wedge q) \leftrightarrow (p \vee \sim q)$
F	T				

Given the true statements p : "A right triangle has one 90-degree angle," q : "The triangle is a right triangle," r : " $a^2 + b^2 = c^2$," and s : "The longest side of a triangle is c implies $a + b$ must be $> c$." Write each of the following compound statements in symbolic form, then construct a truth table to determine the truth value of the compound statement.

14. If a triangle is a right triangle, then it does not have one 90-degree angle or $a^2 + b^2 = c^2$.
15. The triangle is a right triangle, or a right triangle does not have a 90-degree angle, if and only if it is not the case that the longest side of a triangle is c implies $a + b$ must be $> c$.

Use the conditional statement, $p \rightarrow q$: "If Phil Mickelson is 50 years old, then Phil Mickelson won the Player's Championship," to answer the following questions.

16. Write the converse statement in words.
17. If the conditional statement is true, and the hypothesis is true, what is a valid conclusion to the argument?
18. If the conditional statement is true, and the conclusion is false, what is a valid conclusion to the argument?
19. Construct a truth table to analyze all the possible outcomes and determine the validity of the following argument.
 $\sim p \vee q \leftrightarrow q \rightarrow p$
20. Construct a truth table or Venn diagram to prove whether the following argument is valid. If the argument is valid, determine whether it is sound.
If John Mayer played *MTV unplugged*, then some guitars are acoustic. John Mayer played *MTV unplugged*.
Therefore, some guitars are acoustic.

